



ORACLE

OCI Zero Trust Packet Routing

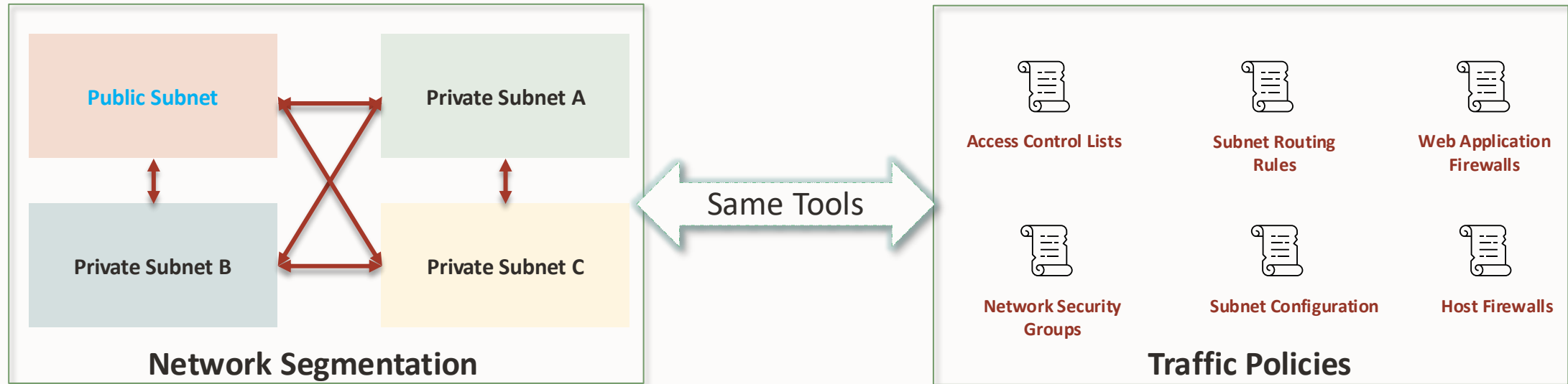
Improved security with reduced effort

Network Architecture and Network Security are not the same

Networking goals: Free flowing communication, resilient architectures, & free of bottlenecks.

VS.

Security goals: Constrain communication between only specific systems and services.

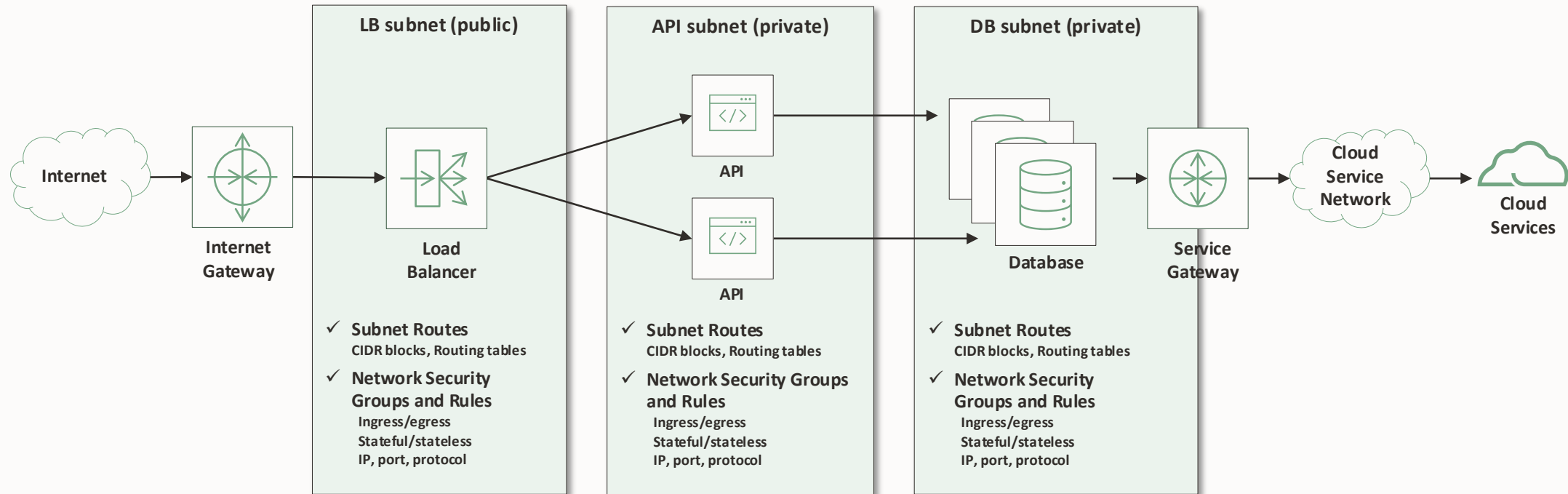


This invariably leads to tradeoffs as architectures evolve over time

- Networks are rendered inflexible by hard and fast security rules on ports or routes
- Security degrades as new ports/routes are opened to fix issues or improve connectivity

Network configuration is complicated by security

Today, we attempt to achieve security via network configuration, which creates complexity.



What should be a simple network and application is **complicated by security**

- **Multiple Subnets** to attempt to build "safe areas" result in more configuration overhead
- **Many Policies** spread across the design
- **Difficult to manage** and maintain the network

Introducing ZPR

A functional overview of ZPR

ZPR delivers Network Security separate from Network Architecture

ZPR delivers the following:

- Solves the problem through separation
- Secures the network using entity identity and ZPR policy and it is implemented in the **network fabric**
- Uses descriptions of security constraints that survive network architecture changes and is evaluated independent of network configuration
- Enables customers to simply describe what can talk to what, using security attributes

Example: in **network:apps** VCN allow **apps:api** endpoints to connect to **apps:database** endpoints

- ZPR policies are independent of network configuration, can be written before, after, or during network configuration.
- Ensures the customer databases will **only ever be reachable** by the API endpoints; irrespective of changes to its IP settings, TCP Ports, move it to a public subnet, etc.

Zero-Trust Packet Routing (ZPR)

Human-Readable Policies

Security attributes and policies designed for management simplicity, not complexity

Clear Trust Boundaries

End-to-end data protection with explicit, enforceable security perimeters

Closes Legacy Gaps

Addresses vulnerabilities that NSGs and routing tables simply cannot protect

Unified Integration

Works seamlessly with PSA and IAM Deny to prevent lateral movement and credential compromise

ZPR In Action: 3 Simple Steps

1

Establish Security Model

2

Deploy ZPR Policies

3

Apply ZPR Attributes



Example Policy in Action

Allow web:app endpoints in network:frontend VCN to connect to customer:db endpoints in network:backend VCN

ZPR console location

Cloud

Search resources, services, documentation, and Marketplace

Australia East (Sydney)

Search

Home

Compute

Storage

Networking

Oracle AI Database

Databases

Analytics & AI

Developer Services

Identity & Security

Observability & Management

Hybrid & Multicloud

Migration & Disaster Recovery

Compartments

Access Governance

Cloud Guard

Overview

Recipes

Alerts

Configuration

Queries

Threat monitoring

Resources

Security Zones

Overview

Zones

Recipes

Security Advisor

Network Firewalls

Network Firewall policies

Web Application Firewall

Policies

Network Address Lists

Edge Policy Resources

Certificates

Certificates

Certificate Authorities

CA Bundles

Scanning

Vulnerability Reports

Scanning Reports

Targets

Scan Recipes

Secret Management

Zero Trust Packet Routing

Overview

Security Attribute Namespace

Policies

Protected Resources

Managed Access

Access Requests

Approval Templates

Resource Settings

Bastion

Compliance

Fri, Feb 27, 2026, 03:11:29 UTC

zpr-events

GET

com.oraclecloud.EventsService.GetRule

200

7

Copyright © 2026, Oracle and/or its affiliates

ZPR User Experience

Customers **create Security Attributes**, **write ZPR policy**, and then **apply Security Attributes** to protected resources.

- Security attributes are used to identify the resources controlled by ZPR policy.
- Security Attributes Namespaces are used to group and authorize Security Attributes.

The screenshot displays the Oracle Cloud console interface. On the left, a sidebar menu for 'Zero Trust Packet Routing' includes links for 'Overview', 'Policies', 'Security Attribute Namespace' (which is highlighted), and 'Protected Resources'. Below the menu, the 'List scope' section shows a compartment dropdown set to 'PM-testing' and a tag filters section with 'add' and 'clear' buttons. The main content area is titled 'Security Attribute Namespace' and contains a 'Create Security Attribute Namespace' button. A modal dialog is open over this, titled 'Create Security Attribute Namespace'. It includes a help section 'What is a Security Attribute Namespace?' explaining that security attributes are used by ZPR to identify resources. The form fields include a 'Compartment' dropdown (set to 'PM-testing'), a 'Name' text input (containing 'Apps'), and a 'Description' text area (containing 'Security Attributes used to describe the applications used by the customer'). A 'Show advanced options' link is present below the description. At the bottom of the dialog are 'Create' and 'Cancel' buttons. The footer of the console shows 'Terms of Use and Privacy' and 'Cookie Preferences' on the left, and a copyright notice 'Copyright © 2024, Oracle and/or its affiliates. All rights reserved.' on the right.



Writing ZPR Policy

ZPR Policies group ZPR policy statements. ZPR policy statements control which Security Attributes can connect to other Security Attributes.

ZPR helps the customer write policy with a “Simple Policy Builder”, “Manual policy builder” and “Policy template builder”.

There are templates for Oracle Database and compute scenarios.

The screenshot shows the Oracle Cloud ZPR Policy Builder interface. At the top, there's a navigation bar with the Oracle Cloud logo, a search bar, and the region 'US Dev West (Seattle)'. The main heading is 'Add policy statements'. Below this, a message states: 'Choose how you want to build your policies.' There are three options: 'Simple policy builder' (Use a wizard to help you construct ZPR policies), 'Manual policy builder' (Enter free-form policy), and 'Policy template builder' (Select a template that provides policy examples based on common use-case scenarios). The 'Policy template builder' is selected. Below the options, there's a 'Category' dropdown set to 'Oracle Database'. Underneath, a 'Choose a policy use case' dropdown is set to 'Select...'. A list of use cases is shown, including 'Enable ExaDB-D service for ssh access, database client access, Object Storage Access, Vault, Data Safe, and other OCI service access, Real Application Clusters (RAC), and Data Guard.', 'Enable BaseDB for all scenarios (includes backup and Data Guard).', 'Allow clients to connect to my database (ExaCS, DBCS, ADBD, ADBS).', 'Provisioning Database/VMCluster.', 'Backup and Restore database to Object Storage.', 'Allow database to access OCI Vault access (KMS).', and 'Allow database to obtain DBaaS metadata.'. Below the use cases, there are three examples of policy statements: 'Example 1: in vcn-network:db VCN allow db-client:app1 endpoints to connect to db-server:app1 endpoints', 'Example 2: in vcn-network:db VCN allow app:front-end endpoints with protocol = 'tcp/999-11199' to connect to app:back-end endpoints', and 'Example 3: in vcn-network:db VCN allow app:front-end endpoints to connect to '192.168.1.1/16' endpoints'. At the bottom, there are 'Add' and 'Cancel' buttons. The footer contains 'Terms of Use and Privacy', 'Cookie Preferences', and 'Copyright © 2024, Oracle and/or its affiliates. All rights reserved.'



Applying Security Attributes to Databases & Compute instances

ZPR makes it easy to apply security attributes by enumerating all resources which may have Security Attributes applied to them.

1. Customers select multiple resources.

1

ORACLE Cloud

Search resources, services, and documentation

US Dev West (Seattle) ▾

⌵

?

🗺

👤

Add security attribute to resources

1 Select resource

2 Add security attributes

3 Review your added resources

Browse available resources to add security attributes to.

Compartment

All

Resource type

All

Select resource

Search

☐	Name	Resource type	OCID	Status	Compartment	Created
☐	beet tmp cinzhao.ad2.zp.r.zpr.zlpr	Vcn	...vue6om2jga Show Copy	● Available	canary_e2e_tests	Mon, Aug 26, 2024, 20:09:39 UTC
☒	beet tmp cinzhao.ad2.zp.r.implicit-rat	Instance	...cuafju4hda Show Copy	● Running	canary_e2e_tests	Mon, Aug 26, 2024, 20:06:58 UTC
☒	beet tmp cinzhao.ad2.zp.r.sharp-salmon	Instance	...vvjqonustq Show Copy	● Running	canary_e2e_tests	Mon, Aug 26, 2024, 20:06:56 UTC
☐	beet tmp cinzhao.ad2.zp.r.mgmt	Vcn	...z4itfwoxra Show Copy	● Available	canary_e2e_tests	Mon, Aug 26, 2024, 20:06:51 UTC
☒	beet tmp cinzhao.ad2.zp.r.gradual-stork	Instance	...b4odkmoxdgq Show Copy	● Terminated	canary_e2e_tests	Mon, Aug 26, 2024, 19:57:25 UTC
☐	beet tmp cinzhao.ad2.zp.r.vertical-squid	Instance	...voghkq2x4q Show Copy	● Terminated	canary_e2e_tests	Mon, Aug 26, 2024, 19:57:24 UTC
☐	beet tmp vasunkri.ad2.zp.r.devoted-panda	Instance	...pfescqvima Show Copy	● Terminated	canary_e2e_tests	Mon, Aug 26, 2024, 19:44:26 UTC
☐	beet tmp vasunkri.ad2.zp.r.fluttering-dragonfly	Instance	...jnlno7orpq Show Copy	● Terminated	canary_e2e_tests	Mon, Aug 26, 2024, 19:44:24 UTC
☐	beet tmp cinzhao.ad2.zo		...gzyydm6pzq Show			

Next

Cancel

Terms of Use and Privacy

Cookie Preferences

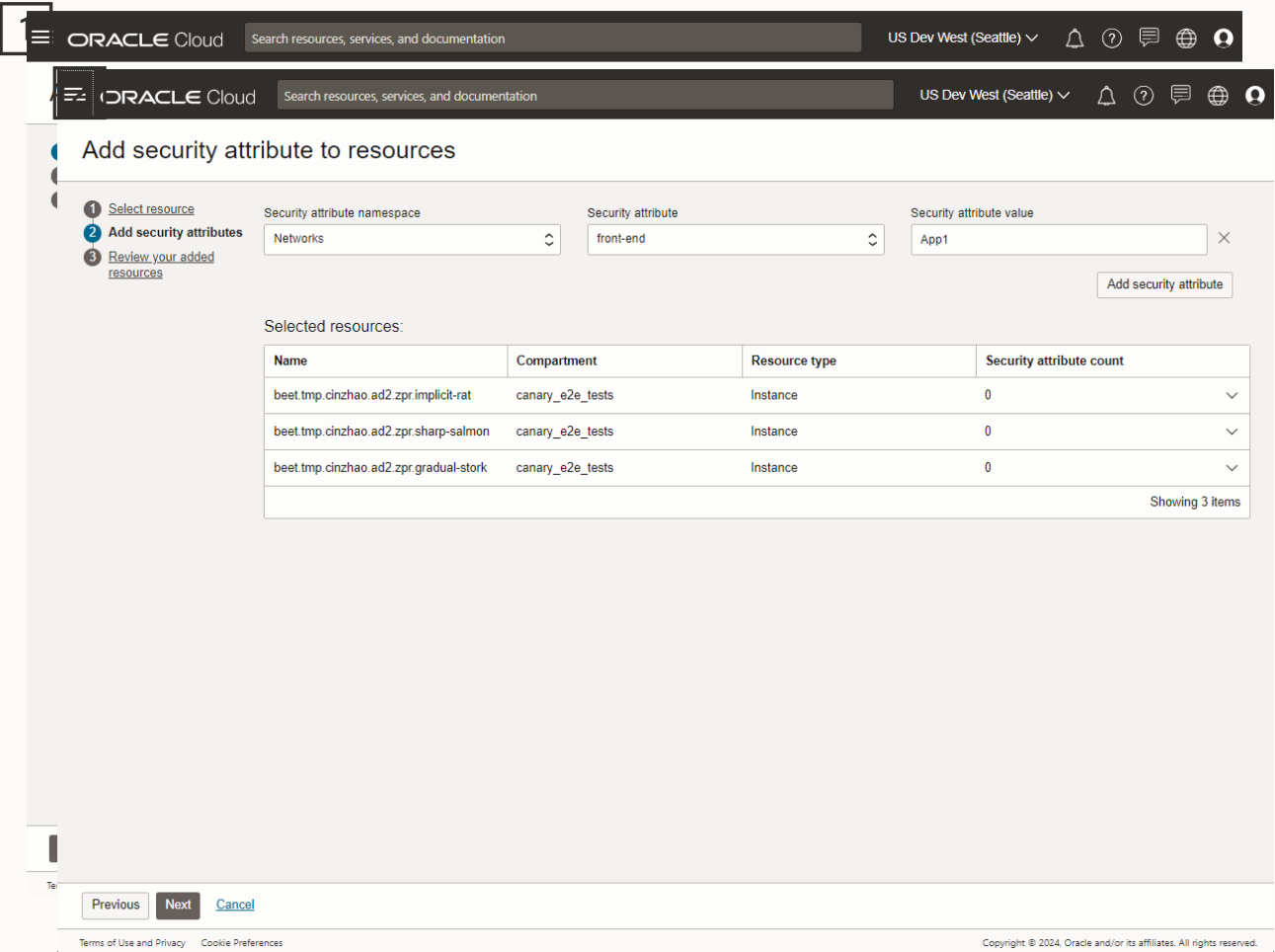
Copyright © 2024, Oracle and/or its affiliates. All rights reserved.



Applying Security Attributes to Databases & Compute instances

ZPR makes it easy to apply security attributes by enumerating all resources which may have Security Attributes applied to them.

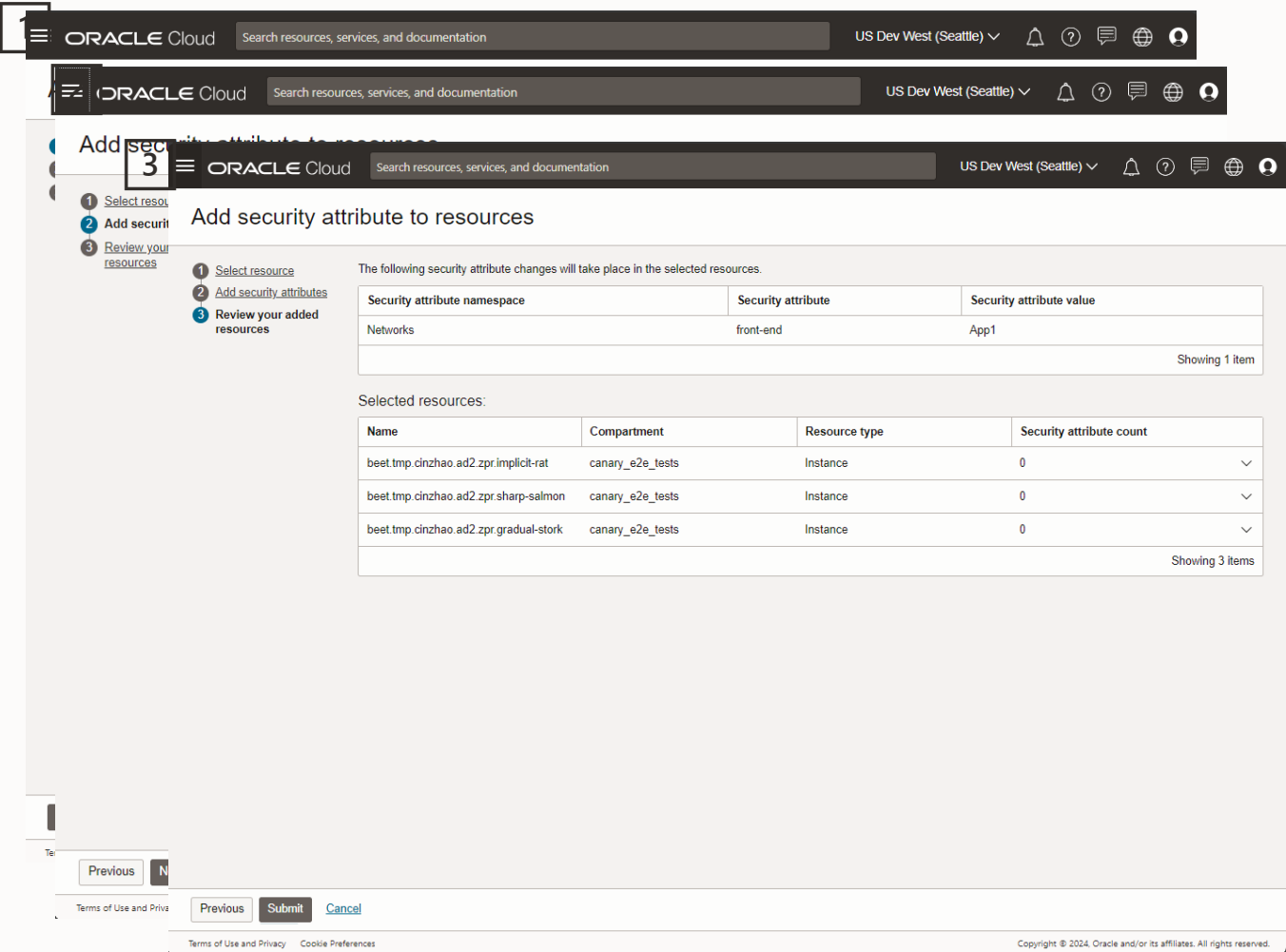
- 1. Customers select multiple resources.
- 2. Choose the Security Attributes to add.



Applying Security Attributes to Databases & Compute instances

ZPR makes it easy to apply security attributes by enumerating all resources which may have Security Attributes applied to them.

- 1. Customers select multiple resources.
- 2. Choose the Security Attributes to add.
- 3. Review and apply the Security Attributes to the resources in bulk.



ZPR Policy Language

ZPR policy language allows customers to describe which computes and databases may connect to each other defined by their security attributes.

```
in <security attribute> vcn allow <security attribute> endpoint to connect to <security attribute> endpoints
```

Example:

```
in network:prod vcn allow apps:prod endpoints to connect to apps:database endpoints
```

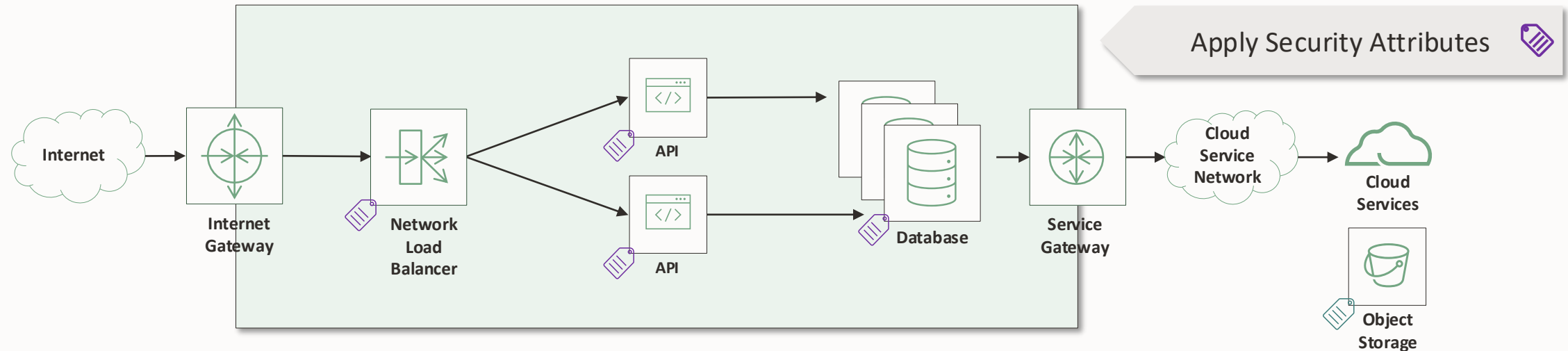
This policy is scoped to the VCN with the security attribute network:prod (key:value). ZPR only allows compute instances with **apps:prod** to connect to databases with **apps:database** security attributes.

You can also specify more precise connections with ports, ip cidrs and protocols.

```
in network:prod vcn allow apps:prod endpoints to connect to apps:database endpoints with protocol=tcp/1521
```

ZPR separates security from network configuration

Managing security outside of network simplifies both security and networking.



in `networks:app1` VCN allow `OCI:LB` endpoints to connect to `Apps:API` endpoints
in `networks:app1` VCN allow `Apps:API` endpoints to connect to `Apps:Database` endpoints
in `networks:app1` VCN allow `Apps:Database` endpoints to connect to `OCI:Object-Storage`

OCI Zero Trust Packet Routing: ZPR Redefines Network Security

Gain simplified control with intent-driven policies that automatically adapt to evolving threats.

The Old Way: IP-Based Chaos

- **Fragile:** Crumbles with network changes
- **Opaque:** Impenetrable, error-prone rules
- **Vulnerable:** Over-permissioned and exposed
- **Lateral Movement:** Attackers roam unchecked
- **Data Exfiltration:** Critical assets at risk

The New Way: Zero Trust Packet Routing

- **Resilient:** Dynamically adapts to change
- **Transparent:** Clear, human-readable policies
- **Secure:** Strict defined clear trust boundaries
- **Containment:** prevents lateral spread instantly
- **Protection:** Safeguards all critical data

With **OCI Zero Trust Packet Routing**, security transforms from a complex, reactive burden into a simple, proactive defense. Simplicity, strength, and sustainable protection, ensuring your most critical assets are protected without the configuration overhead.

Recent Released Feature

Cross VCN Support

Unified policy enforcement across VCNs to simplify management and strengthen Zero Trust Security

Unified Policy Enforcement

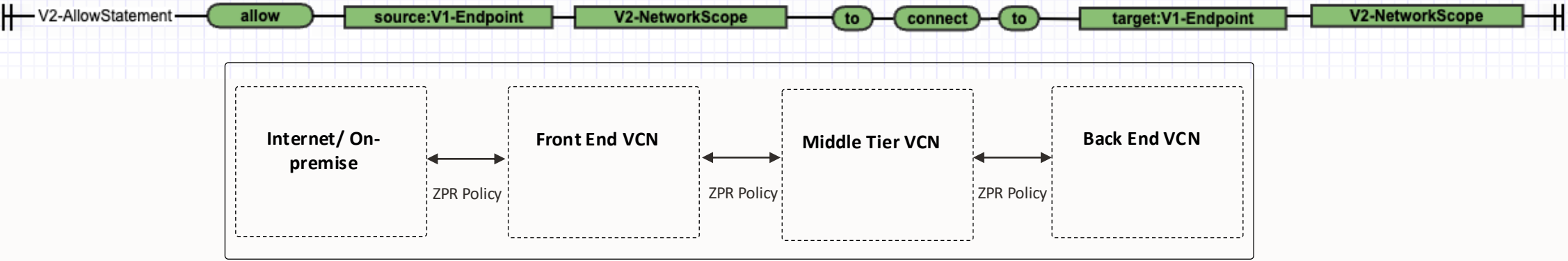
One policy covers workloads in different VCNs - no complex configurations or duplicate policy management required

Strong Zero Trust Posture

Prevents lateral movement and enforces least-privilege access across VCN boundaries

Reduced Complexity

Simplifies security in large-scale, distributed cloud environments.

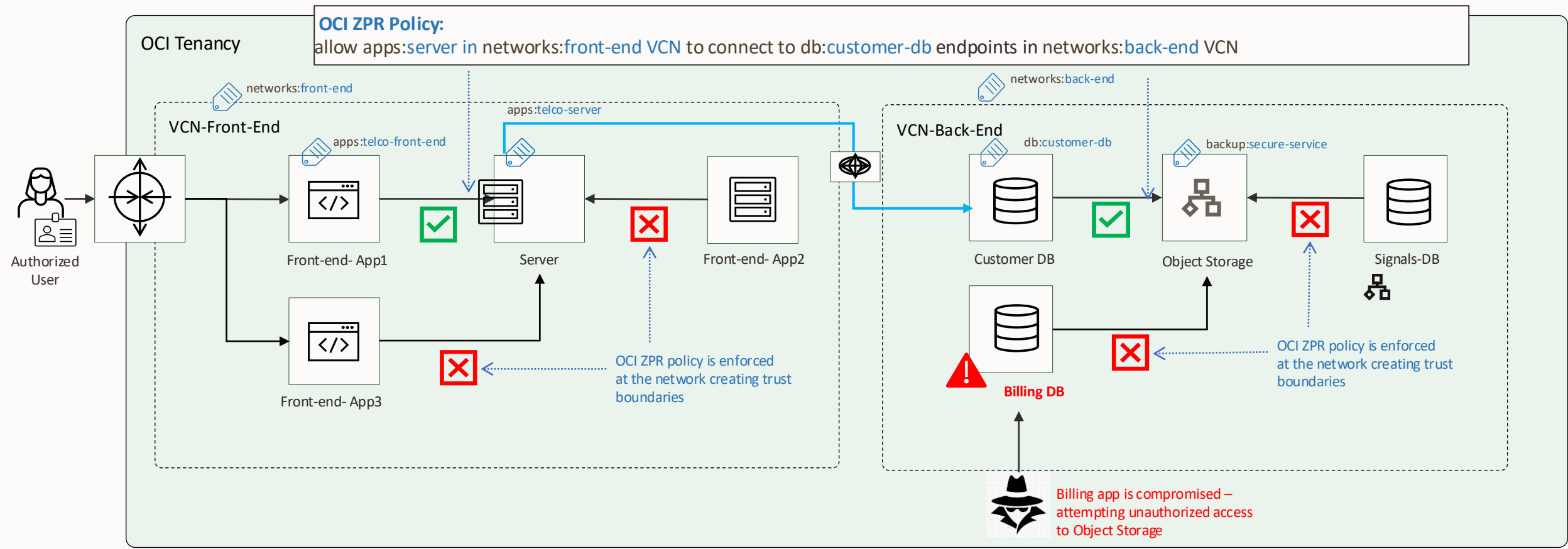


Cross-VCN Policy Syntax

Cross VCN Syntax: Allow `<source attribute>` endpoints in `<source VCN attribute>` VCN to connect to `<destination attribute>` endpoints in `<destination VCN attribute>` VCN



Cross VCN policy support- Example



Flows allowed by ZPR policy — Flows that match ZPR policy based on security attributes and VCN context



Flows blocked by ZPR policy — Flow denied due to missing/incorrect security attributes, unauthorized access attempts



Broadened Coverage for OCI Workloads

Expanding Zero Trust protection across the OCI ecosystem



Initially focused on securing core compute and networking resources:

- Compute Instances
- Exadata CS
- Autonomous Database
- Load Balancer
- Network Load Balancer



Now extending protection to key managed services:

- OCI Resource Manager
- OCI Streaming
- OpenSearch Cluster
- Functions
- HeatWave (MySQL)
- OCI Cache / Redis Clusters
- File Storage Service
- GoldenGate
- Database Tools
- Oracle Integration Cloud
- OCI Network Firewall (API Only)

Integrated ZPR Troubleshooting

Leverage Network Path Analyzer to easily troubleshoot any connectivity issues and identify root cause

With OCI ZPR and Network Path Analyzer, you gain a powerful advantage:

- Seamlessly diagnose with NPA and resolve network configuration issues, pinpointing root causes with ease.

Integrated View

Integrated with Network Path Analyzer for enhanced, end-to-end connectivity troubleshooting of ZPR-protected resources.

Policy Overlay

Overlay ZPR policy decisions directly onto Network Path Analyzer results for a single-pane troubleshooting experience.

Root Cause Analysis

Identify whether blocked connections are due to ZPR policy enforcement or underlying network misconfigurations.

Faster Resolution

Reduce mean time to resolution by quickly pinpointing the exact policy or network element(NSG/SL) blocking traffic.

☰

Cloud

🔍

Search resources, services, documentation, and Marketplace

▼

Australia East (Sydney)

🖥️

🔔

?

👤

🔗 Networking

< Network Command Center

Overview

Network visualizer

Network Path Analyzer

Inter-region latency

Capture filters

VTAPs

Flow logs

Network path analyzer

Network path analyzer helps you identify virtual network configuration issues (up to layer 4) by analyzing the configured connectivity

⚠️ Information

If you are unable to access Network path analyzer, please contact your tenancy-admin for configuration of user policy. [Learn more.](#)

View details

🔍 Search and Filter

Search

Applied filters

Compartment ZPRDemo

Create path analysisDelete

☐☐

☐	Name	Protocol	Source	Source port	Destination	Destination port	Type and code	Created	▼
☐	app-to-db-correct-version	TCP	10.0.1.10	Any	10.0.2.200	1522	—	Feb 23, 2026, 02:54:56 UTC	...
☐	app-to-db	TCP	10.0.1.10	Any	10.0.1.20	1522	—	Feb 23, 2026, 02:51:56 UTC	...

◀▶

Page 1 of 1 (1 - 2 of 2 total items)

Items per page25▼

📄

Terms of Use and Privacy

Cookie Preferences

Copyright © 2026, Oracle and/or its affiliates. All rights reserved.

[Give us feedback](#)

Network Path Analyzer

☰ Cloud

🔍 Search resources, services, documentation, and Marketplace

▼ Australia East (Sydney) 📺 🔔 ⓘ 👤

🔗 Networking

< Network Command Center

Overview

Network visualizer

Network Path Analyzer

Inter-region latency

Capture filters

VTAPs

Flow logs

📄

← Network path analyzer

app-to-db-correct-version Active

Path analysis test

Actions ▾ Analyze

Details Tags

General information

OCID	...c5wx46faixq	Copy
Compartment	ZPRDemo	
Created	Feb 23, 2026, 02:54:56 UTC	
Source type	IP address	
Source IP	10.0.1.10	
Source port	Any	
Protocol	TCP	
Direction	Bi-Directional	
Destination type	IP address	
Destination IP	10.0.2.200	
Destination port	1522	

Discovered paths

Often there is more than one available path for traffic between two points. Network path analyzer can present up to 8 paths between the selected source and destination, but it is possible more could exist. If no complete path is found, the path with the most successful hops is shown.

Analyze

📄

Path	Forward path result	Return path result	Completed date
Press analyze to view analysis results.			

Terms of Use and Privacy Cookie Preferences

Copyright © 2026, Oracle and/or its affiliates. All rights reserved.

[Give us feedback](#)

Path analysis result

Executed date Mon, 23 Feb 2026 03:10:19 UTC

Forward path result

Status Reachable

Successful hops 1

Path hop	From	To	Routing status	Security status	Traffic
▶ Hop 1	DEMO_APP_1	10.0.2.200	Forwarded	Allowed	TCP: 10.0.1.10:49152 to 10.0.2.200:1522

◀

▶

Page 1 of 1 (1 - 1 of 1 total items)

Items per page 25 ▼

Return path result

Status Reachable

Successful hops 1

Path hop	From	To	Routing status	Security status	Traffic
▶ Hop 1	10.0.2.200	DEMO_APP_1	Forwarded	Allowed	TCP: 10.0.2.200:1522 to 10.0.1.10:49152

◀

▶

Page 1 of 1 (1 - 1 of 1 total items)

Items per page 25 ▼

Network Path Analyzer

Path hop	From	To	Routing status	Security status	Traffic
▼ Hop 1	DEMO_APP_1	10.0.2.200	Forwarded	Allowed	TCP: 10.0.1.10:49152 to 10.0.2.200:1522
✔ Routing forwarded Forwarded based on implicit routing. ✔ ZPR Allowed ZPR security attributes and policies are configured to allow network traffic between the source and destination endpoints. View policies and statements ✔ Ingress access control: Security allowed Allowed based on rule: [all protocols any port from 10.0.0.0/16] found in the following security list: Default Security List for VCN-DEMO ✔ Egress access control: Security allowed Allowed based on rule: [all protocols to 0.0.0.0/0 : any port] found in the following security list: Default Security List for VCN-DEMO					

Hop from	
Subnet	subnet-middle
VCN	VCN-DEMO
VNIC	ocid1.vnic.oc1.ap-sydney-1.abzxsljrwd635j6fdbqk3fbtsns7ukk43un4kxgpu4g7y5vs47pm45dquyq
Instance	DEMO_APP_1
Hop to	
VCN	VCN-DEMO
Subnet	subnet-db
VNIC	lp:10.0.2.200:ocid1.vnic.oc1.ap-sydney-1.abzxsljrbttfoyseop4stcj2rr3vhpwdyzirlxj4zqawm3s4fcxpkggxhtaa

Network Path Analyzer

ChromeFileEditViewHistoryBookmarksProfilesTabWindowHelp

HomeRulesLog CLog CLog CLog CAuditSubsDashLog EHomeDashDashLog ELog ELog E

cloud.oracle.com/networking/path-analyzer?region=ap-sydney-1

StarShareGoogle TranslatePrintDownload

Cloud

Search resources, services, documentation, and Marketplace

Australia East (Sydney)

Path analysis result

Path hop	From	To	Routing status	Security status	Traffic
▼ Segment	DEMO_APP_1	DEMO_APP_2	No route	Denied	TCP: 10.0.1.10:49152 to 10.0.1.20:1522

No route

We don't have information about the routing for this segment of the path. There could be multiple hops, or it could be completely unroutable.

Blocked - No ZPR Policy Permits this Path

ZPR security attributes are present on both the source and destination endpoints. Relevant ZPR policies are not configured to permit network traffic between these endpoints.

View policies and statements

Ingress access control: Security not applicable

Security information is not applicable

Egress access control: Security allowed

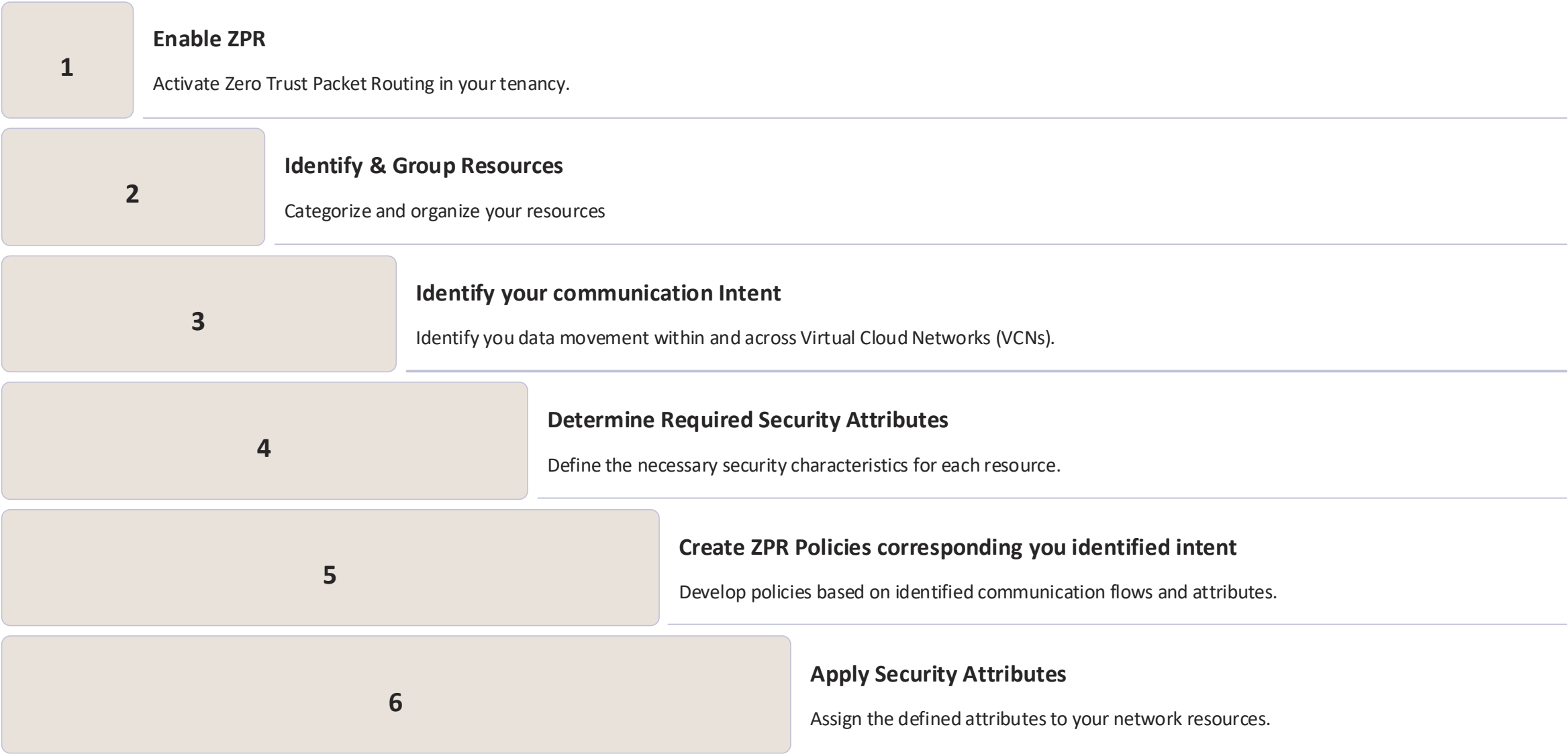
Allowed based on rule: [all protocols to 0.0.0.0/0 : any port] found in the following security list: [Default Security List for VCN-DEMO](#)

Segment from

Close

25 Copyright © 2026, Oracle and/or its affiliates

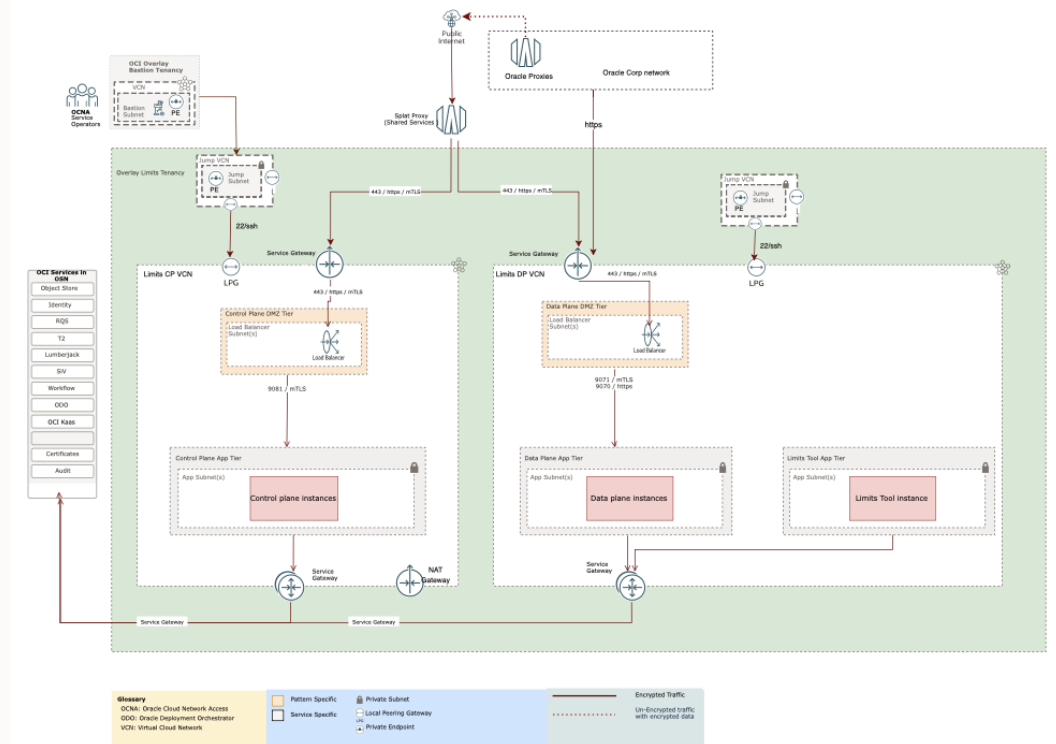
Adopting ZPR: From Existing Controls to ZPR Controls



ZPR Adoption

Understand the existing network controls

- List your Networking Resources – VCNs, Subnets, Security Lists, NSGs, Gateways
- List all your dependencies and Create PSA for each



ZPR Adoption

Trust Boundary	Security-Attribute
VCN	zpr-tools-stage.network:cp
Load Balancers	<u>zpr-tools-stage.lb</u> :cp
Application Servers	zpr-tools-stage.app:cp
Worker Servers	zpr-tools-stage.worker:cp
Private Service Access (PSA)	zpr-tools-stage.psa:storage
Bastion	zpr-tools-stage.bastion:cp



ZPR Adoption Key Benefits

- **Intent-based security:** Network access is defined by Security Attributes (app/lb/bastion/psa groups) instead of subnet/CIDR-specific Security Lists—policies follow workloads as they scale, move, or get replaced.
- **Stronger segmentation:** ZPR expresses explicit source → destination flows (e.g., LB→App, Bastion→App:22, Scan→targets), reducing blast radius and limiting lateral movement.
- **Improved auditability & operations:** Human-readable policies make it easier to validate “who can talk to whom,” speed up security reviews, and simplify troubleshooting/change management.
- **Private service enforcement:** PSA enables private connectivity to supported OCI services; IAM Deny blocks access via non-PSA gateways, reducing bypass paths and helping prevent unintended public-path access/exfiltration.



Common Misconfigurations

- Orphaned resources not assigned security-attributes
- Possible mismatches between security-attributes on resources and ZPR policy statements
- Applying security-attributes before creating ZPR policies
- Applying security-attributes on endpoints before applying on VCN



FAQ

- **Does ZPR get enabled automatically?**

No. ZPR must be explicitly enabled for your tenancy or selected compartments.

- **Once ZPR is enabled, are Security Attributes applied automatically?**

No. Security Attributes must be explicitly applied to resources before ZPR policies take effect.

- **What happens when a Security Attribute is applied to a resource?**

The resource's communication is evaluated based on ZPR policies. Existing NSG and Security List rules continue to apply unless fully transitioned to ZPR-only.

- **How does ZPR interact with NSGs and Security Lists?**

ZPR coexists with NSGs and Security Lists. Traffic is allowed only if permitted by both; if either denies, the traffic is dropped.

- **Do we need to migrate all workloads to ZPR at once?**

No. Migration can be phased. You can start with selected VCNs or compartments while others remain under NSG/SL control.

- **During migration, can ZPR policies use IP addresses?**

Yes. Transition policies can reference IP addresses to maintain connectivity and avoid downtime.

- **Will there be any downtime during migration?**

No. Using transition IP-based policies allows seamless migration without downtime.

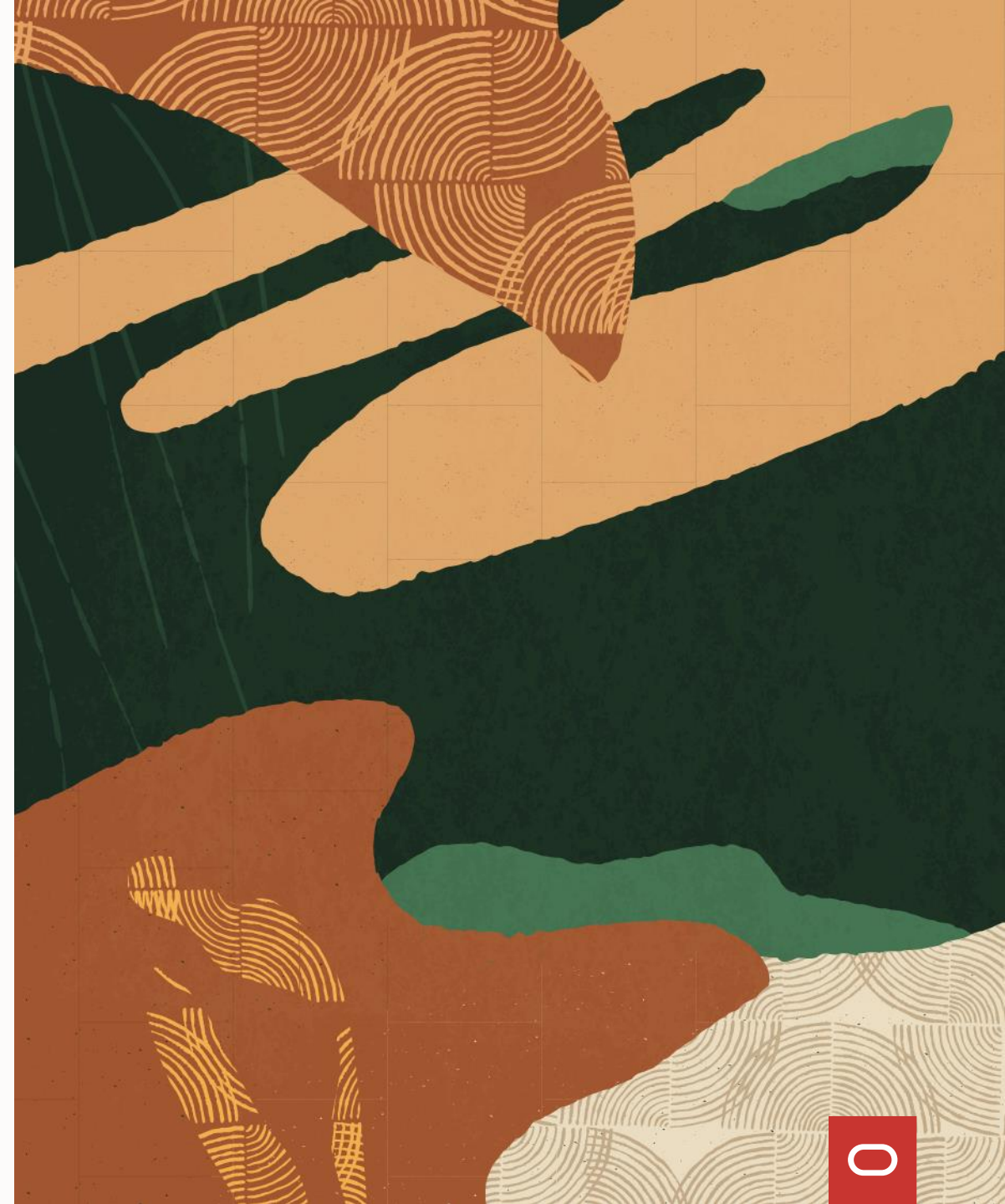


Thank you



Learn More

oracle.com/security



ORACLE